Circular Ceiling Diffuser RCD/RCD-R/RCDH/RCD-RH



RCD-R



RCDH

Description
Ceiling diffusers with circular or square front panel and adjustable circular opening.

Application
Supply/ exhaust of the air flow.

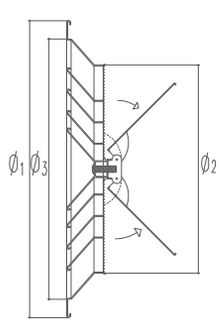
Characteristics
Recommended difference between air temperature and indoor air is ± 10 K .
Conic shape with removable central core.
Depending on the type of roof and front panel, the following alternatives are possible:
RCD - for low ceiling with circular front panel
RCD-R - for low ceiling with square front panel 595 X 595 mm.
RCDH - for high ceiling with circular front panel
RCD-RH - for high ceiling with square front panel 595 X 595 mm.

Accessories
Insulated/Non-insulated adapter **ADP IZ/N**

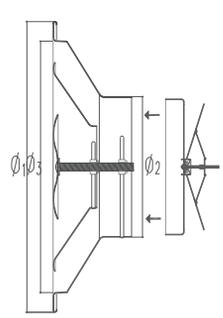
Materials and finishing
Aluminum material electrostatic field painted white RAL 9016.
Other RAL colors available on request.

Technical design

RCD



RCDH



Available dimensions

RCDH / RCD / RCD-RH / RCD-R			
Diffuser type	Ø2	Ø1	Ø3
RCDH	100	250	170
	160	335	280
	200	423	360
	250	517	445
	315	640	555
	355	730	640
	400	776	680
	450	825	735
	500	915	820
	630	1045	943
RCD	150	258	208
	200	308	255
	250	358	305
	300	408	355
	350	458	405
RCD-RH	160	□595X595	280
	200		360
	250		445
	315		555
RCD-R	200	□595X595	395
	250		395
	300		395

Selection table RCD/RCD-R/RCDH/RCD-RH

RCDH/RCD-RH - high ceiling

Air flow m³/h	Connection size Ø(mm)	160	200	250	315	355	400	450	500	630
	Ak [m²]	0.015	0.027	0.04	0.056	0.074	0.09	0.011	0.013	0.019
200	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]	1.4 25.0 3.7 14.0								
250	X [m] NR [dB(A)] Veffm/s] ΔPt [pa]	1.7 30.0 4.4 20.0	1.3 18.0 2.5 6.6							
300	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]	2.1 36.0 5.6 33.0	1.5 23.0 3.0 10.0	1.3 15.0 2.0 4.2						
400	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]	2.8 45 7.2 54	2.1 30.0 4.0 17.0	1.7 22.0 2.7 7.7	1.5 3.0 2.0 4.2					
500	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]		2.6 36.0 5.0 26.0	2.1 27.0 3.4 12.0	1.8 20.0 2.4 6.0					
600	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]		3.2 42.0 6.0 38.0	2.6 33.0 4.0 17.0	2.2 25.0 3.0 10.0	1.9 20.0 2.3 5.0				
800	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]			3.5 41.0 5.6 33.0	3.0 34.0 4.0 17.0	2.6 25.0 3.0 9.0	2.3 20.0 2.5 6.5			
1000	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]			4.5 50.0 7.0 52.0	3.7 40.0 5.0 26.0	3.3 34.0 3.8 16.0	3.0 28.0 3.0 9.0	2.7 22.0 2.5 6.5		
1250	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]				4.6 47.0 6.2 40.0	4.1 42.0 4.8 23.0	3.8 35.0 3.8 16.0	3.3 26.0 3.0 9.0	3.0 24.0 2.6 6.9	
1500	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]					5.0 47.0 5.5 32.0	4.5 41.0 4.6 22.0	4.0 35.0 3.8 16.0	3.6 29.0 3.1 10.0	
1750	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]							4.8 40.0 4.4 20.0	4.3 34.0 3.6 14.0	3.5 25.0 2.6 6.9
2000	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]								4.9 38.0 4.1 19.0	4.1 30.0 2.9 8.0
2500	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]								6.0 45.0 5.0 29.0	5.1 37.0 3.6 14.0

RCD/RCD-R - low ceiling

Air flow m³/h	Connection size Ø(mm)	160	200	250	315	355
	Ak [m²]	0.011	0.02	0.031	0.046	0.047
150	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]	1.8 - 3.9 1.0	1.3 - 2.1 6.0			
200	X [m] NR [dB(A)] Veffm/s] ΔPt [pa]	2.3 27.0 5.0 22.0	1.8 - 2.8 7.8	1.4 - 1.9 -		
300	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]	3.4 41.0 7.5 50.0	2.6 23.0 4.1 15.0	2.1 - 2.7 7.2	1.7 - 1.9 -	
400	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]		3.4 33.0 5.7 28.0	2.7 20.0 3.6 12.0	2.3 - 2.5 5.5	1.9 7.0 2.4 3.2
500	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]		4.3 41.0 7.0 45.0	3.5 27.0 4.5 18.0	2.8 15.0 3.1 8.0	2.0 10.0 2.8 4.8
600	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]			4.1 34.0 5.5 26.0	3.3 22.0 3.7 14.0	2.8 20.0 3.5 7.0
800	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]			5.5 45.0 7.5 50.0	4.6 32.0 5.0 22.0	3.5 30.0 4.8 12.0
1000	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]				5.7 40.0 6.2 35.0	4.5 37.0 6.0 20.0
1200	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]				6.7 46.0 7.5 50.0	5.4 40.0 7.0 28.0
1400	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]					7.1 50.0 10.2 39.2
1600	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]					7.8 57.0 11.6 50.3

Correction coefficients for other velocities inside sitting area.

Vk[m/s]	0.2	0.25	0.375	0.5
x[m]	1.25	1	0.75	0.5

Annotation

* Ak [m²] - Free surface
* Veff[m/s] - Air velocity inside the diffuser
* ΔPt [pa] - Pressure loss
* X [m] - Air jet length
* NR [dB(A)] - Noise level without the room attenuation
* Vk = 0.25 [m/s] - velocity at which the air jet length was calculated "x" [m]

Order codes

Type	+	ADP IZ/N	/	Dimensions	/	0
				Ø		
RCDH/RCD **RCD-RH/RCD-R**		Insulated/ Non-insulated adapter		high ceiling: 160, 200, 250, 315, 355, 400, 450, 500, 630 low ceiling: 160, 200, 250, 315, 355		**0** Painted white RAL 9016 **R** Other RAL paints